

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 17

ATMOSPHERIC PRESSURE

CORIOLIS FORCE

FACTORS AFFECTING THE WIND VELOCITY

PRESSURE GRADIENT FORCE

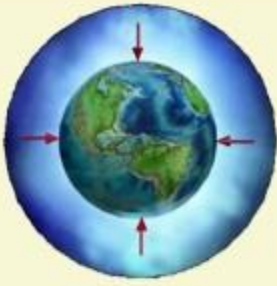
The rate of change of pressure with respect to distance is the pressure gradient. The pressure gradient results in a net force that is directed from high to low pressure and this force is called the "pressure gradient force".

FRICTIONAL FORCE

It affects the speed of the wind. It is greatest at the surface and its influence generally extends upto an elevation of 1–3km. Over the seasurface the friction is minimal.

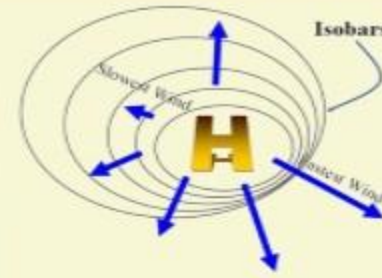
CORIOLIS FORCE

The rotation of the earth about its axis affects the direction of the wind. It is a deflective force and it deflects the wind to the right direction in the northern hemisphere and to its left in the southern hemisphere.



ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)



1 The coriolis force is a **pseudo force** that **deflects any moving object** like wind, ocean currents, flying birds, aircrafts, ballistic missiles.

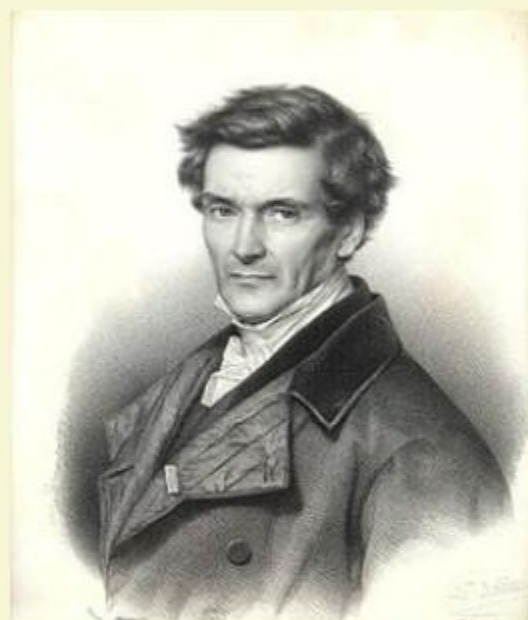
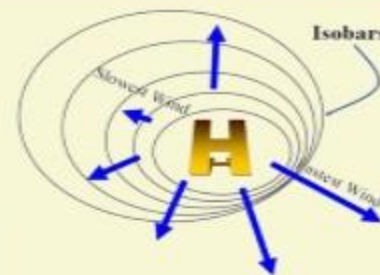
2 It is **a not a force** rather it is the effect of rotational movement of the Earth.

Coriolis Effect : Original path of air is deflected westward by the rotation of the planet

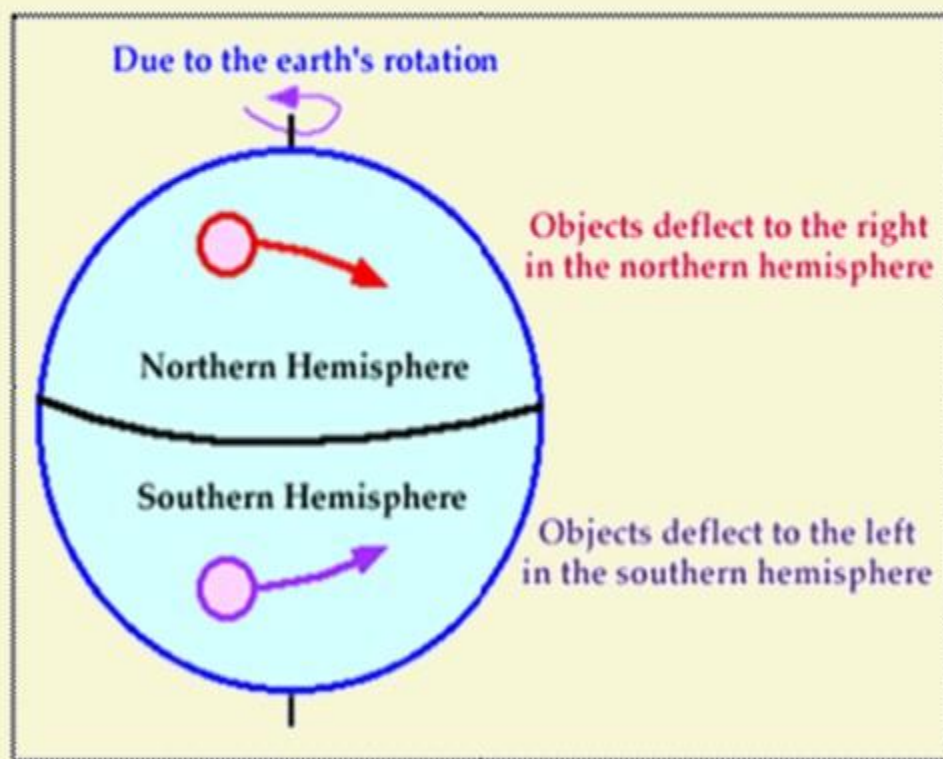


ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)

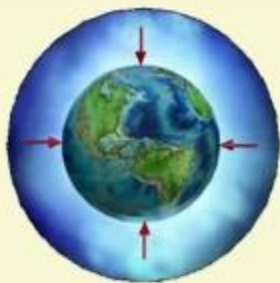


G.G. Coriolis



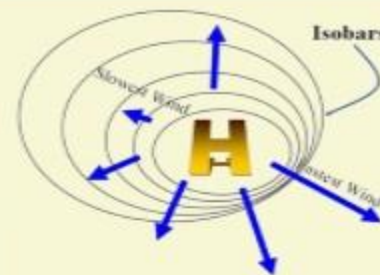
1 Gaspard-Gustave Coriolis was the one who described the Coriolis Force back in 1835.

2 Because of CF, all winds are deflected to the right in the northern hemisphere while they are deflected to the left in the southern hemisphere.



ATMOSPHERIC PRESSURE

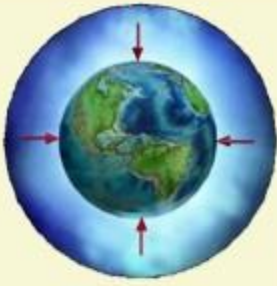
CORIOLIS FORCE(EFFECT)



The red ball always travels in a straight line but the "track" of the red ball over the spinning globe makes it appear to loop around the North Pole.

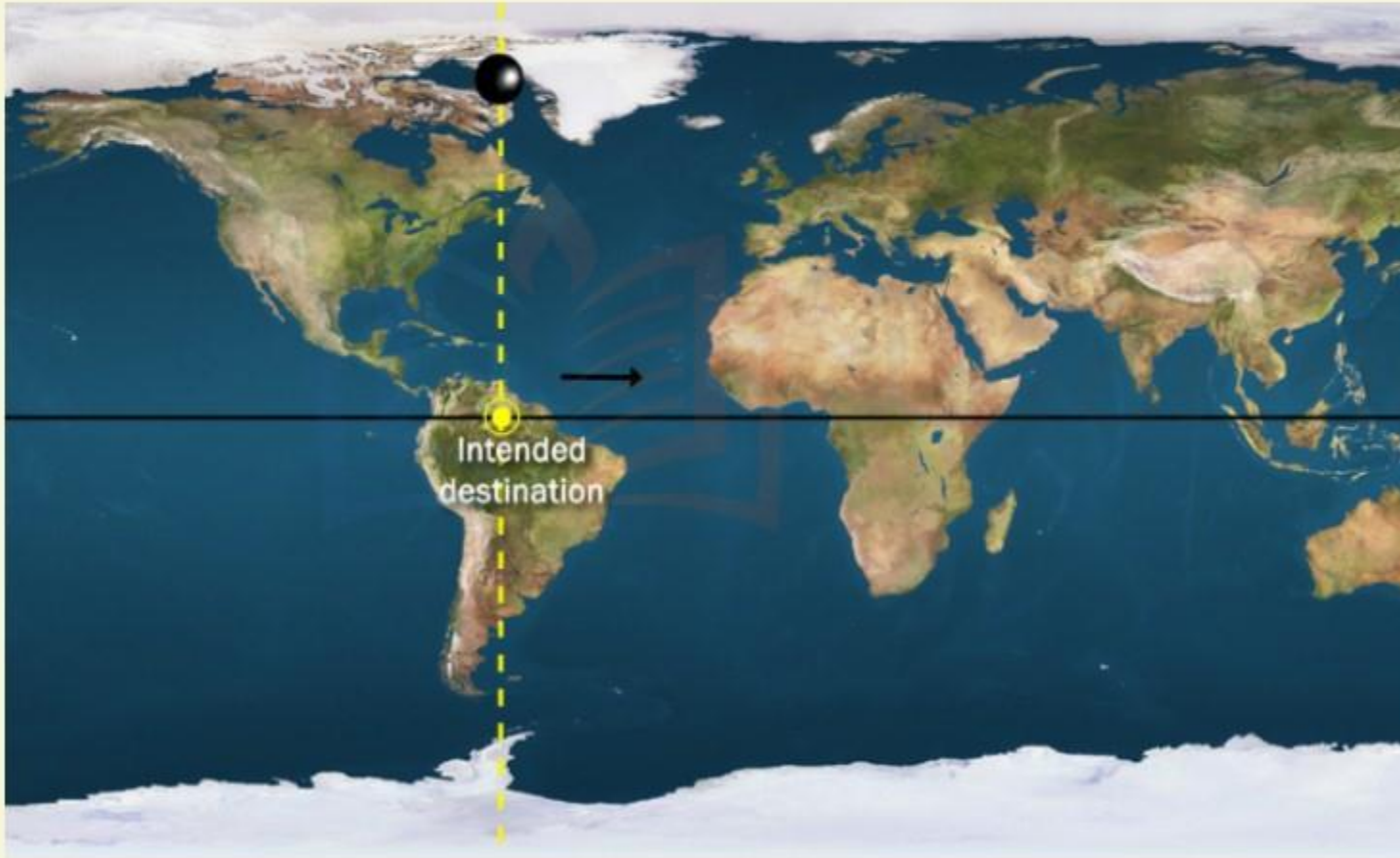
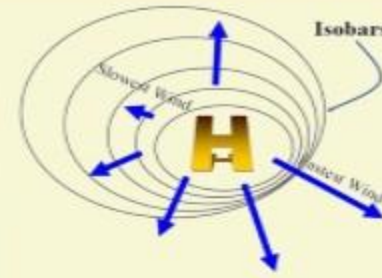
HOW DOES CF WORK IN REALITY?

The red ball always travel in a straight line but because of the Earth's rotation the ball seems to be looping over its actual path.



ATMOSPHERIC PRESSURE

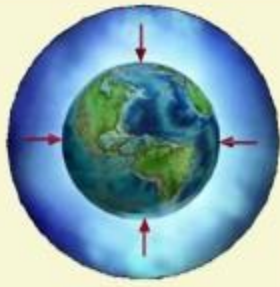
CORIOLIS FORCE(EFFECT)



HOW DOES CF WORK IN REALITY?

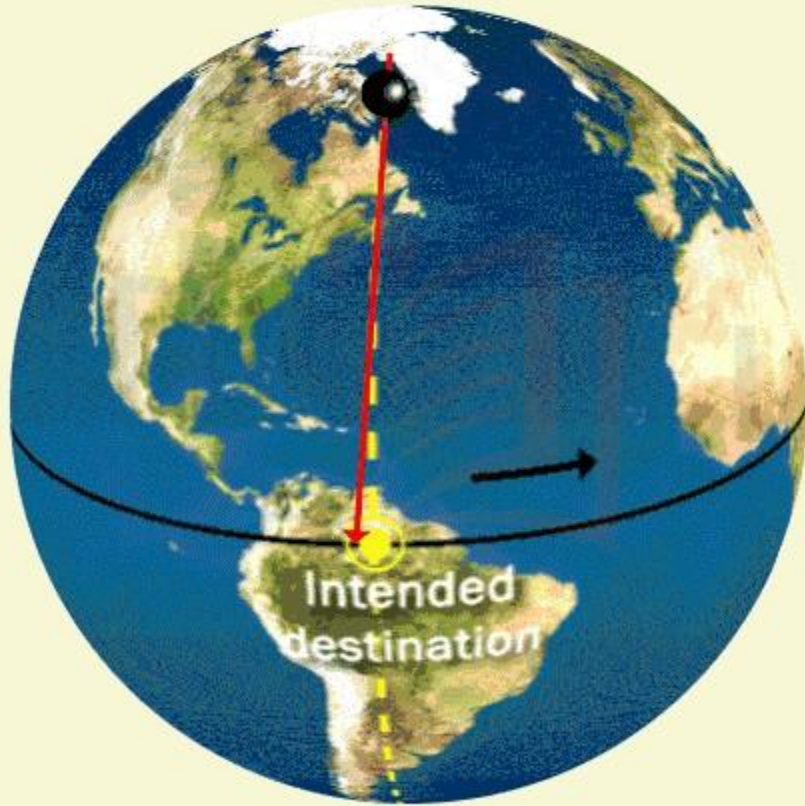
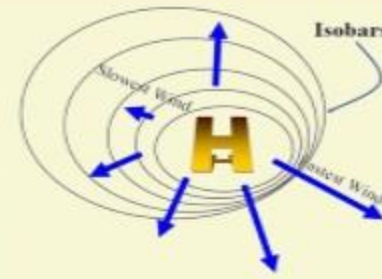
- 1 Let us assume that we are shooting a cannon ball from Greenland to South America.





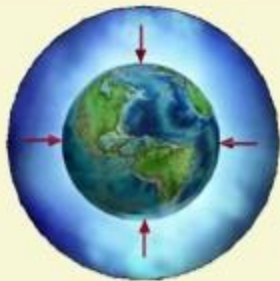
ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)



HOW DOES CF WORK IN REALITY?

- 2 The cannon is rotating along with the earth to the east and thus along with its initial speed it is **getting the additional speed of rotating earth**.
- 3 However, as the cannon moves equatorwards, it flies over land points whose linear velocity is much **higher**. This is because the equator spins much faster than the poles.



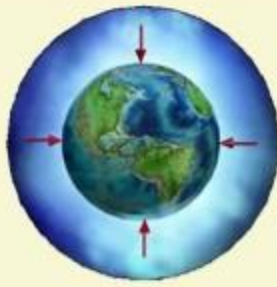
ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)



HOW DOES CF WORK IN REALITY?

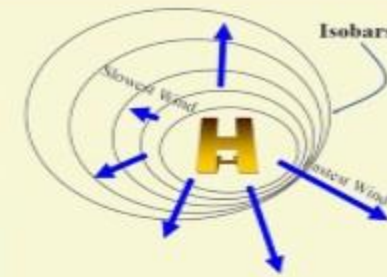
- 4 The low inertia of the cannon ball causes it to land westwards, that is, behind its intended destination.
- 5 Thus, the cannon will fall on a meridian located to the west of the one from which it was fired, although the direction of the shot was exactly to the south.
- 6 Thus, because of the CF the canon ball will seem to be moving its right in the northern hemisphere.



ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)

CHARACTERISTICS OF CORIOLIS FORCE



**Important concept
for prelims/mains**

- 1** Every latitude is a complete circle. Equatorial latitudinal circle is the largest one and latitudinal circles decrease poleward wherein polar circle is smallest one.
- 2** The whole Earth completes one rotation along its axis roughly in 24 hours. Thus, the rotational speed of the Earth is highest at the equator and decreases poleward.
- 3** When the wind moves towards the equator following straight path, it does not reach its destination because by that time the destination place moves ahead and the wind lags behind because of high rotational speed of the Earth.
- 4** Contrary to this the wind moving either northward or southward in high latitudes reaches ahead of its destination because of decreasing rotational speed of the Earth as one moves towards the poles.



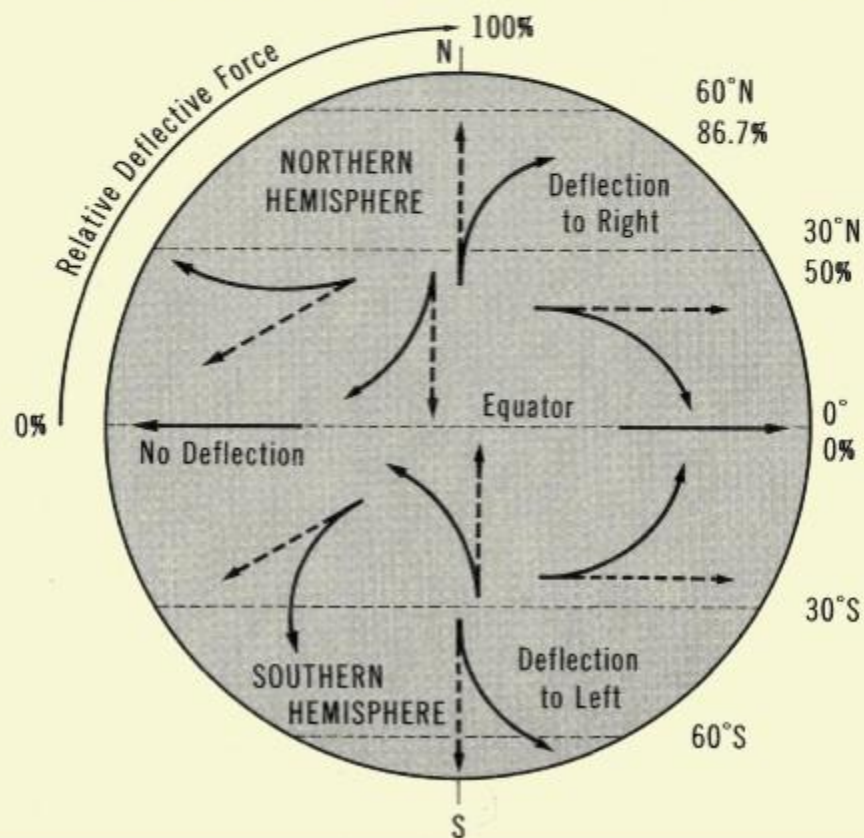
ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)

CHARACTERISTICS OF CORIOLIS FORCE



DEFLECTIVE FORCE OF EARTH'S ROTATION



5 CF affects wind direction and not wind speed.

6 However, its magnitude is determined by wind speed.

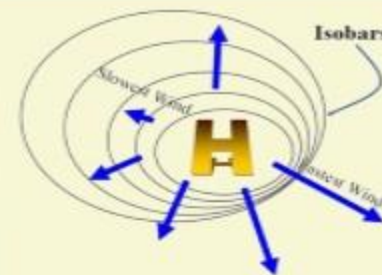
7 The higher the wind speed, the greater is the deflection of the wind direction due to resultant greater deflective force and vice versa.



ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)

CHARACTERISTICS OF CORIOLIS FORCE

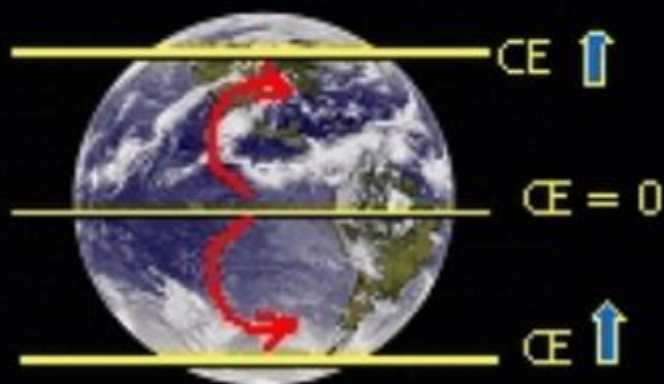


Coriolis Effect: Summary

occurs to:
the right in NH

Equator

the left in SH



CE ↑

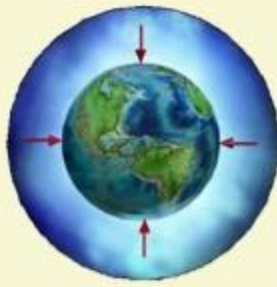
CE = 0

CE ↑

(Increases with latitude)

8

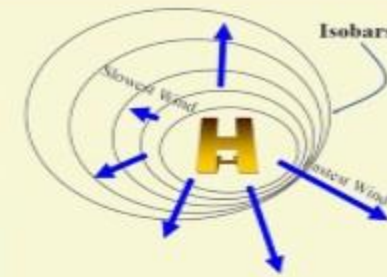
CF becomes maximum at the poles due to minimum rotational speed of the earth while it becomes zero at the equator.



ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)

CHARACTERISTICS OF CORIOLIS FORCE

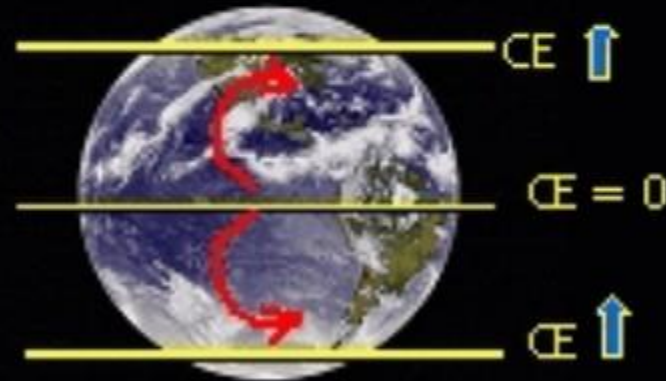


Coriolis Effect: Summary

occurs to:
the right in NH

Equator

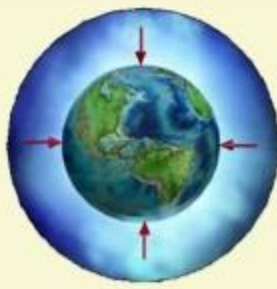
the left in SH



(Increases with latitude)

9 The magnitude of deflection is directly proportional to

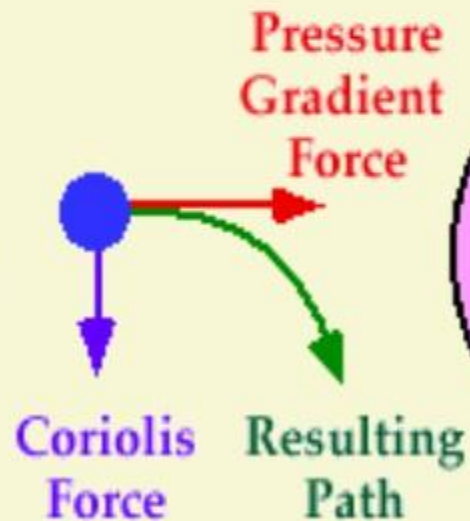
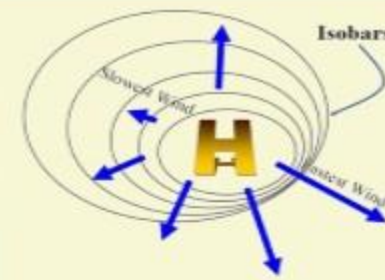
- The **sine of the latitude** ($\sin 0^\circ = 0$, $90^\circ = 1$)
- The **mass** of the moving body
- **Horizontal velocity of the wind.**



ATMOSPHERIC PRESSURE

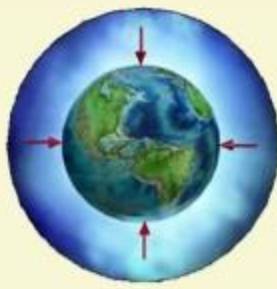
CORIOLIS FORCE(EFFECT)

CHARACTERISTICS OF CORIOLIS FORCE



10 It acts always at right angles to the horizontally moving air and other moving objects.

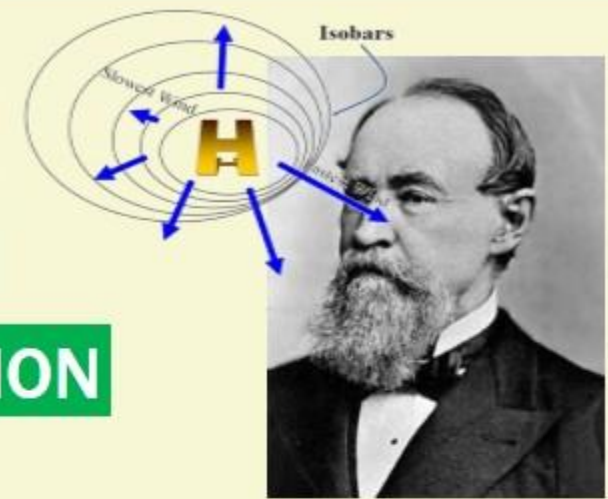
11 The net effect is that the horizontal winds are deflected to the right in the northern hemisphere and to the left in the southern hemisphere.



ATMOSPHERIC PRESSURE

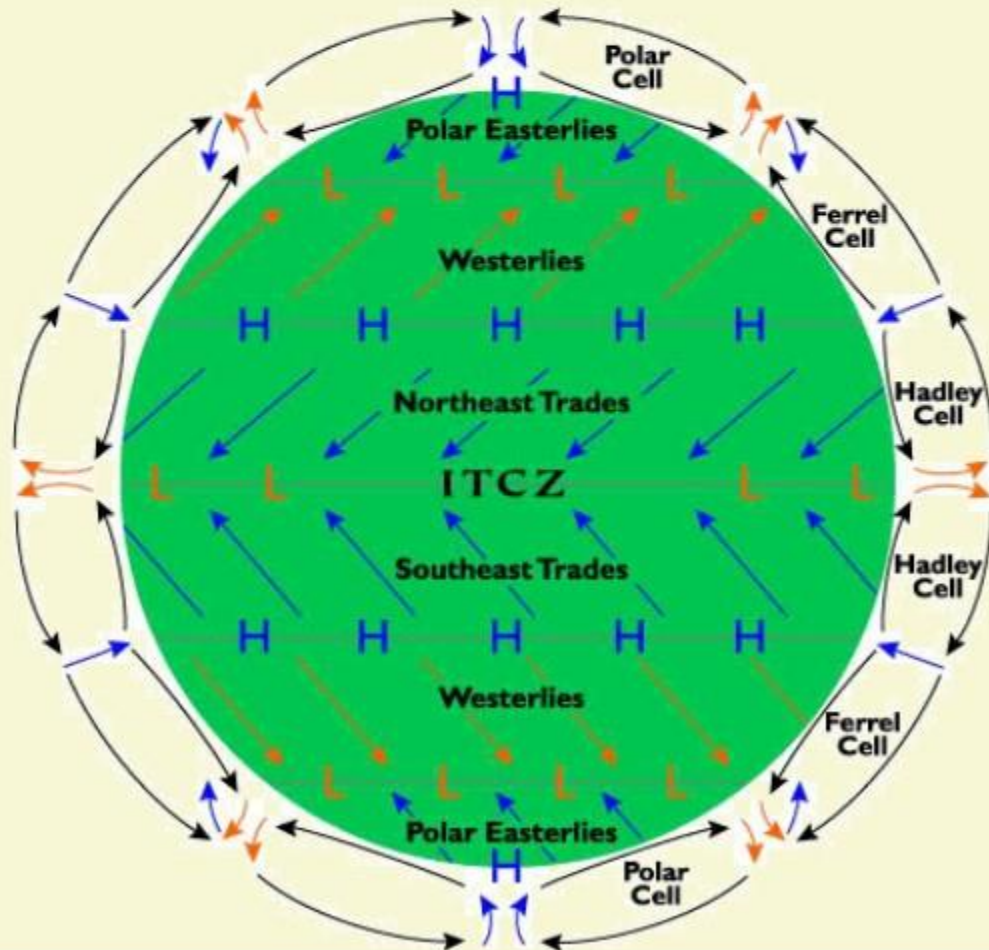
CORIOLIS FORCE(EFFECT)

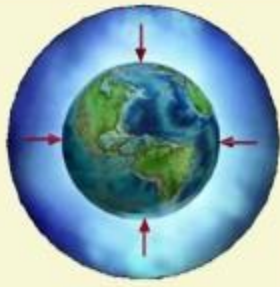
LAWS GOVERNING THE WIND DIRECTION



FERREL'S LAW

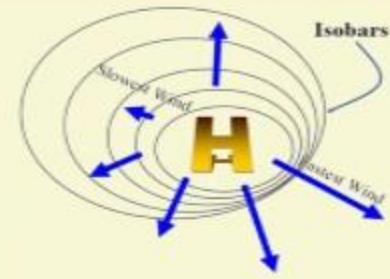
- 1 Based on the deflective force (Coriolis force) of the Earth and deflection of winds, Ferrel has propounded his law which is popularly known as **Ferrel's law**.
- 2 This law states that 'if one stands with one's back towards the direction from where winds are coming, they (winds) are deflected to the right in the northern hemisphere and to the left in the southern hemisphere.'





ATMOSPHERIC PRESSURE

CORIOLIS FORCE(EFFECT)

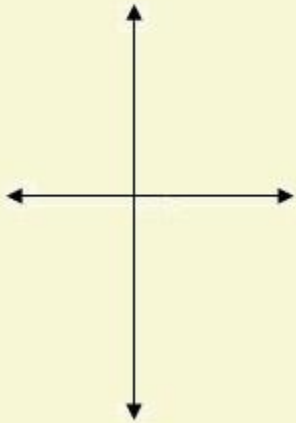


LAWS GOVERNING THE WIND DIRECTION

NAMING OF WINDS

Winds are named for the *direction from which they flow* and not the direction where they are moving towards.

North



E.g. West Wind



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 - 2 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9